

ERSS: Final Project

Protocol Specification

Interoperability Group:3

team members:

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```

// port: 34567
syntax = "proto2";
// Amazon -> UPS
message ACommand {
    repeated AOrderTruck toOrder=1;
    repeated AStartDelivery toStart=2;
    repeated ACheckUsername checkUsers=3;
    repeated int64 acks=4;
}

//One package can contain multiple product, so change it to repeated
message AOrderTruck {
    required int64 packageID = 1;
    required product productInfo=2;
    repeated product productInfo=2;
    required warehouse warehouseInfo=3;
    required destination destinationInfo=4;
    required string upsUsername = 5;
    required int64 seqnum=6;
}

message product{
    required int64 productID= 1;
    required string description=2;
    required int32 count = 3;
}

message warehouse{
    required int32 warehouseID = 1;
    required int32 x = 2;
    required int32 y = 3;
}

message destination {
    required int32 x=1;
    required int32 y=2;
}

message AStartDelivery {
    required int64 packageID = 1;
    required int64 seqnum=2;
}

message ACheckUsername {
    required string upsUsername=1;
    required int64 seqnum=2;
}

```

```
}
```

```
// UPS -> Amazon
```

```
message UCommand {  
    repeated UTruckArrival arrived=1;  
    repeated UDelivered delivered=2;  
    repeated UCheckUsernameResponse checkUser=3;  
    repeated Err error=4;  
    repeated int64 acks=5;  
}
```

```
// UCommand add a new command: UChangeDestination  
since If UPS change destination, it should send new destination to Amazon
```

```
message UCommand {  
    repeated UTruckArrival arrived=1;  
    repeated UDelivered delivered=2;  
    repeated UCheckUsernameResponse checkUser=3;  
    repeated UChangeDestination changed=4;  
    repeated Err error=5;  
    repeated int64 acks=6;  
}
```

```
// error message
```

```
message Err{  
    required string err = 1;  
    required int64 originseqnum = 2;  
    required int64 seqnum = 3;  
}
```

```
message UTruckArrival {  
    required int64 packageID = 1;  
    required int32 truckID=2;  
    required int64 seqnum=3;  
}
```

```
message UDelivered {  
    required int64 packageID = 1;  
    required int64 seqnum=2;  
}
```

```
message UCheckUsernameResponse {  
    required string upsUsername=1;
```

```

    //-1: username does not exist
    required int64 upsUserID=2;
    required int64 seqnum=3;
}

// If UPS change destination, it should send new destination to Amazon
message UChangeDestination {
    required int64 packageID = 1;
    required destination NewDestination=2;
    required int64 seqnum=3;
}

```

1. Each request **MUST** have a seqnum.
2. All the Amazon to UPS request **SHOULD** be wrapped within ACommand, and all UPS to Amazon request **SHOULD** be wrapped within UCommand.
3. When Amazon receives an order, Amazon **SHOULD** send AOrderATruck to UPS to order a truck.
4. When truck arrived at warehouse, UPS **SHOULD** send UTruckArrival to Amazon.
5. When package is loaded, Amazon **SHOULD** send AStartDelivery to UPS.
6. When package is delivered, UPS **SHOULD** send Udelivered to Amazon.
7. When Amazon want to specify UPS account name of order, Amazon **MAY** send ACheckUsername to UPS, and UPS send UCheckUsernameResponse to Amazon.
8. When change destination, UPS **SHOULD** send UChangeDestination to Amazon.